MA7442 Assessment 1 - Task 1

Anonymous

The Dataset

Read in Packages and data

```
library(tidyverse)
Financial <- read.csv("Data/dataset1.csv")
```

Overview

The dataset chosen contains financial and employment infromation from adults, extracted from the US 1994 Census database (Dua and Graff, 2019). It includes 48842 observations of 15 demographic and employement-related attributes. For example, the dataset contains variables related to age, education, workclass, hours worked per week, and income levels. Using this data can help provide insights into job security, wage gaps, and workforce stability, however it is important to note that the social and economic context from 1994 may not reflect current realities and potentially reinforce historical inequalities. Therefore, it is important that any models used do not amplify stereotypes or disadvantage marginalised groups.

Objectives of Exploratory Data Analysis

The key objectives of this EDA are:

- To understand the distribution of key demographic and employment attributes
- To explore potential relationships between variables
- To identify patterns that could be useful for modeling tasks

Exploratory Data Analysis

Explore Dataset

When exploring the dataset, it is found that missing values are presented as either " " or "?", therefore these missing indicators were converted and standardised to NA in order to count the missing values in each column.

```r
#Display first 10 rows
head(Financial, 10)
```

```
   age        workclass fnlwgt education education.num        marital.status
1   39        State-gov  77516 Bachelors            13         Never-married
2   50 Self-emp-not-inc  83311 Bachelors            13    Married-civ-spouse
3   38          Private 215646   HS-grad             9              Divorced
4   53          Private 234721      11th             7    Married-civ-spouse
5   28          Private 338409 Bachelors            13    Married-civ-spouse
6   37          Private 284582   Masters            14    Married-civ-spouse
7   49          Private 160187       9th             5 Married-spouse-absent
8   52 Self-emp-not-inc 209642   HS-grad             9    Married-civ-spouse
9   31          Private  45781   Masters            14         Never-married
10  42          Private 159449 Bachelors            13    Married-civ-spouse
          occupation  relationship  race    sex capital.gain capital.loss
1       Adm-clerical Not-in-family White   Male         2174            0
2    Exec-managerial       Husband White   Male            0            0
3  Handlers-cleaners Not-in-family White   Male            0            0
4  Handlers-cleaners       Husband Black   Male            0            0
5     Prof-specialty          Wife Black Female            0            0
6    Exec-managerial          Wife White Female            0            0
7      Other-service Not-in-family Black Female            0            0
8    Exec-managerial       Husband White   Male            0            0
9     Prof-specialty Not-in-family White Female        14084            0
10   Exec-managerial       Husband White   Male         5178            0
   hours.per.week native.country income
1              40  United-States  <=50K
2              13  United-States  <=50K
3              40  United-States  <=50K
4              40  United-States  <=50K
5              40           Cuba  <=50K
6              40  United-States  <=50K
7              16        Jamaica  <=50K
8              45  United-States   >50K
```

```
9               50  United-States   >50K
10              40  United-States   >50K
```

```r
#Count missing values in each column
Financial[Financial == "" | Financial == "?"] <- NA
colSums(is.na(Financial))
```

```
           age      workclass         fnlwgt      education  education.num 
             0           2799              0              0              0 
marital.status     occupation   relationship           race            sex 
             0           2809              0              0              0 
  capital.gain   capital.loss hours.per.week native.country         income 
             0              0              0            857              0 
```

```r
#Display summary statistics
summary(Financial)
```

```
      age         workclass             fnlwgt         education         
 Min.   :17.00   Length:48842       Min.   :  12285   Length:48842       
 1st Qu.:28.00   Class :character   1st Qu.: 117550   Class :character   
 Median :37.00   Mode  :character   Median : 178144   Mode  :character   
 Mean   :38.64                      Mean   : 189664                      
 3rd Qu.:48.00                      3rd Qu.: 237642                      
 Max.   :90.00                      Max.   :1490400                      
 education.num   marital.status      occupation        relationship       
 Min.   : 1.00   Length:48842       Length:48842       Length:48842       
 1st Qu.: 9.00   Class :character   Class :character   Class :character   
 Median :10.00   Mode  :character   Mode  :character   Mode  :character   
 Mean   :10.08                                                            
 3rd Qu.:12.00                                                            
 Max.   :16.00                                                            
     race               sex             capital.gain    capital.loss     
 Length:48842       Length:48842       Min.   :    0   Min.   :   0.0   
 Class :character   Class :character   1st Qu.:    0   1st Qu.:   0.0   
 Mode  :character   Mode  :character   Median :    0   Median :   0.0   
                                       Mean   : 1079   Mean   :  87.5   
                                       3rd Qu.:    0   3rd Qu.:   0.0   
                                       Max.   :99999   Max.   :4356.0   
 hours.per.week  native.country        income            
 Min.   : 1.00   Length:48842       Length:48842       
 1st Qu.:40.00   Class :character   Class :character   
```

```
 Median :40.00   Mode  :character   Mode  :character
 Mean   :40.42
 3rd Qu.:45.00
 Max.   :99.00
```

Data Cleaning and prep

- Overall 6465 missing values were found and omitted from the dataset to facilitate data visualisation and compatibility with future modelling methods.
- Within the dataset, the income variable has inconsistent labels, therefore these were renamed.
- The following irrelevant columns were dropped:
  - 'Capital.gain' and 'Capital.loss' removed as they introduced high variance
  - 'fnlwgt' Removed as it is a census adjustment therefore not a meaningful predictor
- All categorical variables were converted into factors for further analysis and compatibility with modelling techniques.
- Rows containing the 'Without-pay' result in the work class variable were removed as a key target variable is 'income', therefore it is irrelevant and distorts the data.

```r
# Handling missing values
Financial_clean <- na.omit(Financial)

# renaming values
Financial_clean <- Financial_clean %>%
  mutate(income = case_when(
    income == ">50K." ~ ">50K",
    income == "<=50K." ~ "<=50K",
    TRUE ~ income
  ))

# dropping irrelevant columns
Financial_clean <- Financial_clean |>
  select(-c(capital.gain, capital.loss, fnlwgt))

# Converted categorical variables to factors where required
str(Financial_clean)
```

```
'data.frame':   45222 obs. of  12 variables:
 $ age           : int  39 50 38 53 28 37 49 52 31 42 ...
 $ workclass     : chr  "State-gov" "Self-emp-not-inc" "Private" "Private" ...
 $ education     : chr  "Bachelors" "Bachelors" "HS-grad" "11th" ...
 $ education.num : int  13 13 9 7 13 14 5 9 14 13 ...
 $ marital.status: chr  "Never-married" "Married-civ-spouse" "Divorced" "Married-civ-spouse"
 $ occupation    : chr  "Adm-clerical" "Exec-managerial" "Handlers-cleaners" "Handlers-clean
 $ relationship  : chr  "Not-in-family" "Husband" "Not-in-family" "Husband" ...
 $ race          : chr  "White" "White" "White" "Black" ...
 $ sex           : chr  "Male" "Male" "Male" "Male" ...
 $ hours.per.week: int  40 13 40 40 40 40 16 45 50 40 ...
 $ native.country: chr  "United-States" "United-States" "United-States" "United-States" ...
 $ income        : chr  "<=50K" "<=50K" "<=50K" "<=50K" ...
 - attr(*, "na.action")= 'omit' Named int [1:3620] 15 28 39 52 62 70 78 94 107 129 ...
  ..- attr(*, "names")= chr [1:3620] "15" "28" "39" "52" ...
```

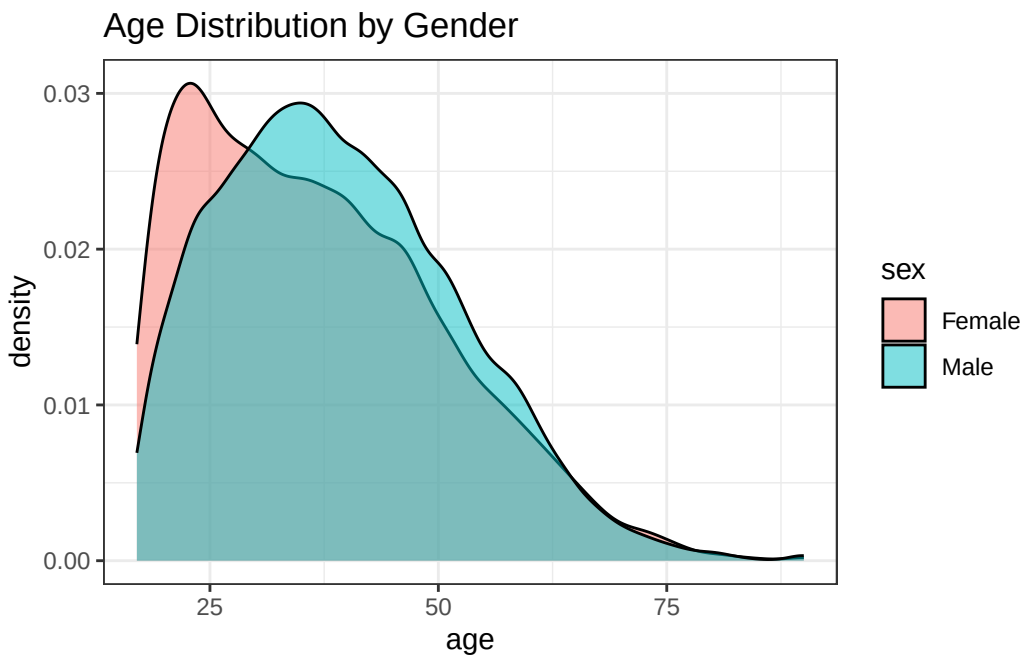
```r
Financial_clean$workclass <- as.factor(Financial_clean$workclass)
Financial_clean$education <- as.factor(Financial_clean$education)
Financial_clean$marital.status <- as.factor(Financial_clean$marital.status)
Financial_clean$occupation <- as.factor(Financial_clean$occupation)
Financial_clean$relationship <- as.factor(Financial_clean$relationship)
Financial_clean$race <- as.factor(Financial_clean$race)
Financial_clean$sex <- as.factor(Financial_clean$sex)
Financial_clean$native.country <- as.factor(Financial_clean$native.country)
Financial_clean$income <- as.factor(Financial_clean$income)

#Removing "without-pay" workclass as looking at target variable income
Financial_clean <- Financial_clean |>
  filter(workclass != "Without-pay")
```

Visualisations

Distribution of key Demographics

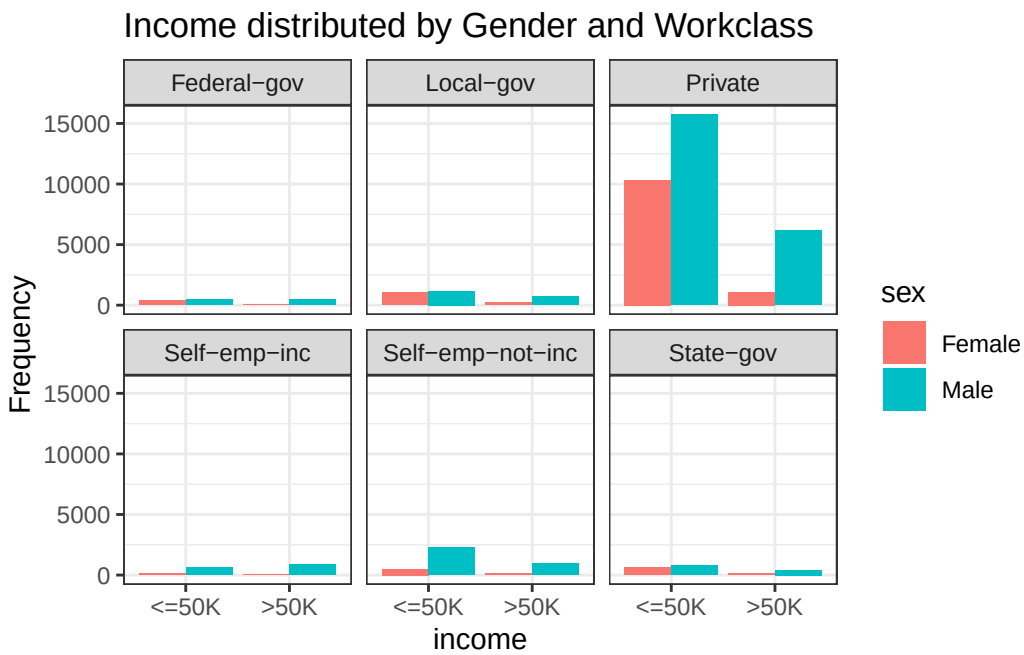
```r
# Distribution of Age by gender
ggplot(Financial_clean, aes(x = age, fill = sex)) +
  geom_density(alpha = 0.5) +
  labs(title = "Age Distribution by Gender") +
  theme_bw()
```



The density plot allows for easy comparison of the distribution of ages between genders within the dataset, highlighting age-related trends or disparities. There are no noticeable differences in age distribution between the genders, both female and male having similar peaks, however the mean for males tends to be around 10 years higher than females.

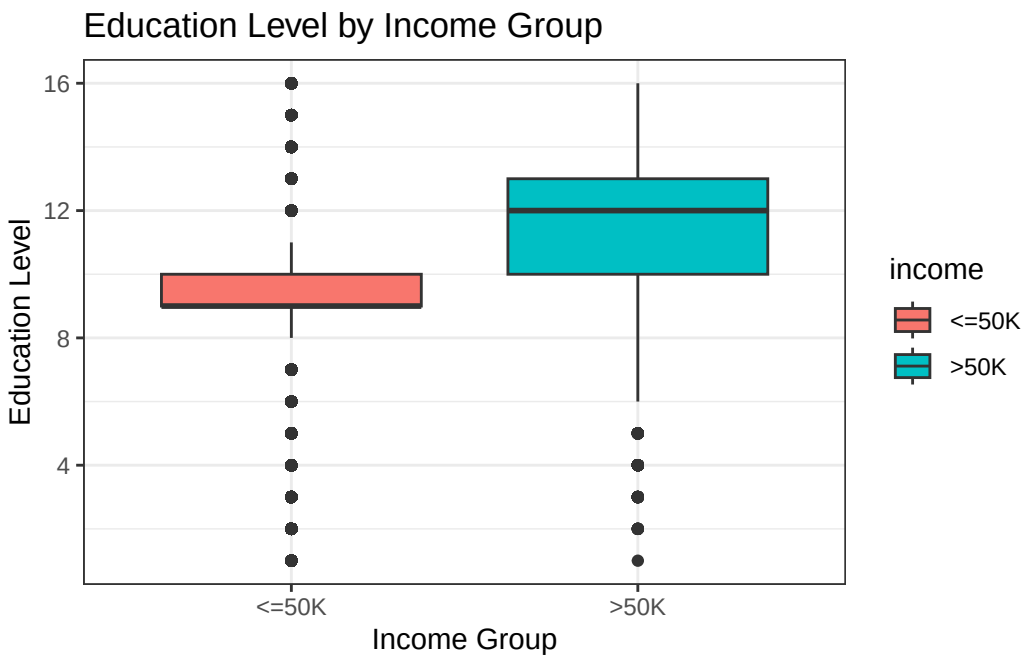
EDA of Income Variable

```
# Income distributed by gender and Workclass
ggplot(Financial_clean, aes(income, fill = sex)) +
  geom_bar(position = "dodge") +
  facet_wrap(~workclass) +
  labs(title = "Income distributed by Gender and Workclass",
       y = "Frequency") +
  theme_bw()
```



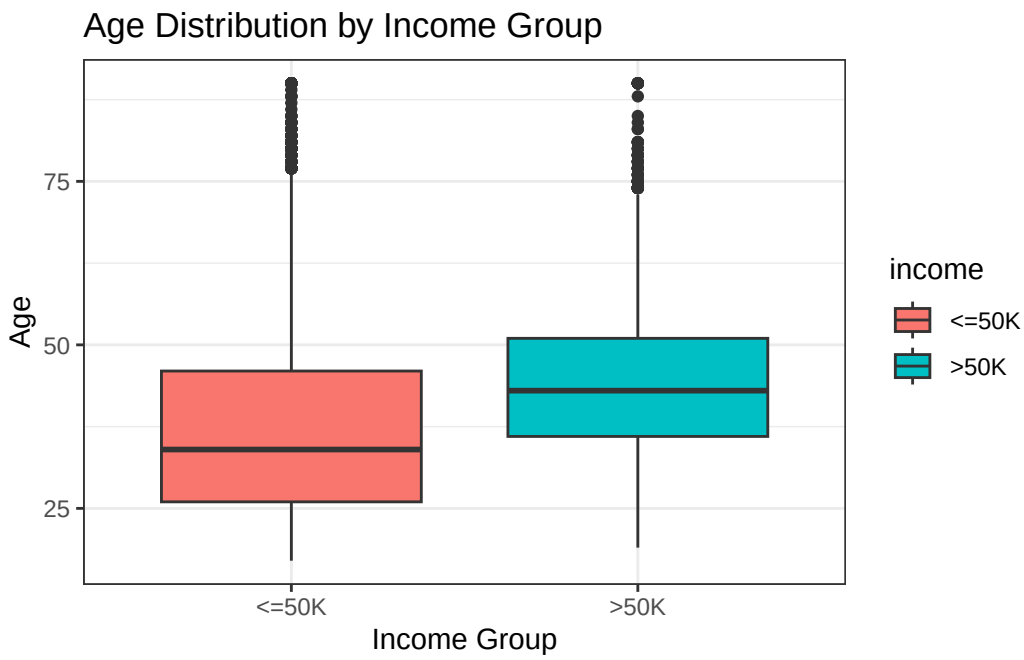
This visualisation allows us to examine how income is distributed by gender within different work classes, allowing us to analyse patterns of income inequality or gender disparities in different sectors. The plot clearly shows that a majority of people work within the private Work class and that overall males tend to earn a higher income. However, its important to note that a disadvantage of this bar chart is the use of count instead of proportion, meaning visible differences could be due to a higher population of males.

```
# Boxplot of education level by income category
ggplot(Financial_clean, aes(x = income, y = education.num, fill = income)) +
  geom_boxplot() +
  labs(title = "Education Level by Income Group",
       x = "Income Group",
       y = "Education Level") +
  theme_bw()
```



This boxplot helps visualise the relationship between education and income by showing the spread and central tendency of education levels within each income group. This boxplot highlights that education is a critical factor in influencing income, with the median for the higher income group being at education 12, whilst being at 9 for <=50K. The spread of education levels in the >50k income group is also much larger, showing that there is a larger variety in education levels at higher income groups- but those education levels still remain higher.

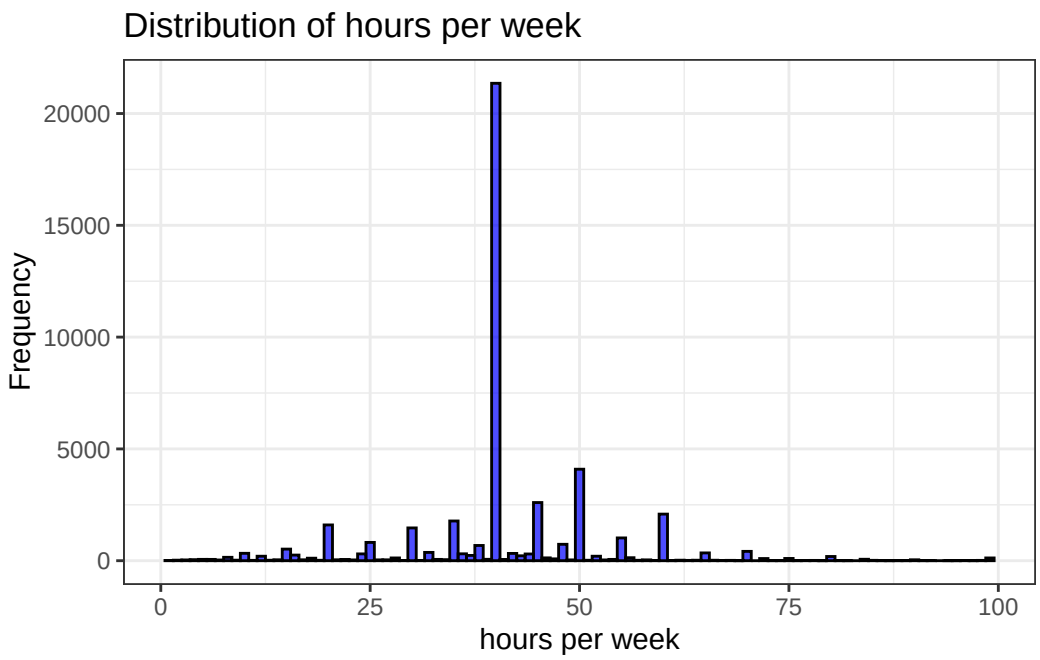
```
# Boxplot of age by income category
ggplot(Financial_clean, aes(x = income, y = age, fill = income)) +
  geom_boxplot() +
  labs(title = "Age Distribution by Income Group",
       x = "Income Group",
       y = "Age") +
  theme_bw()
```



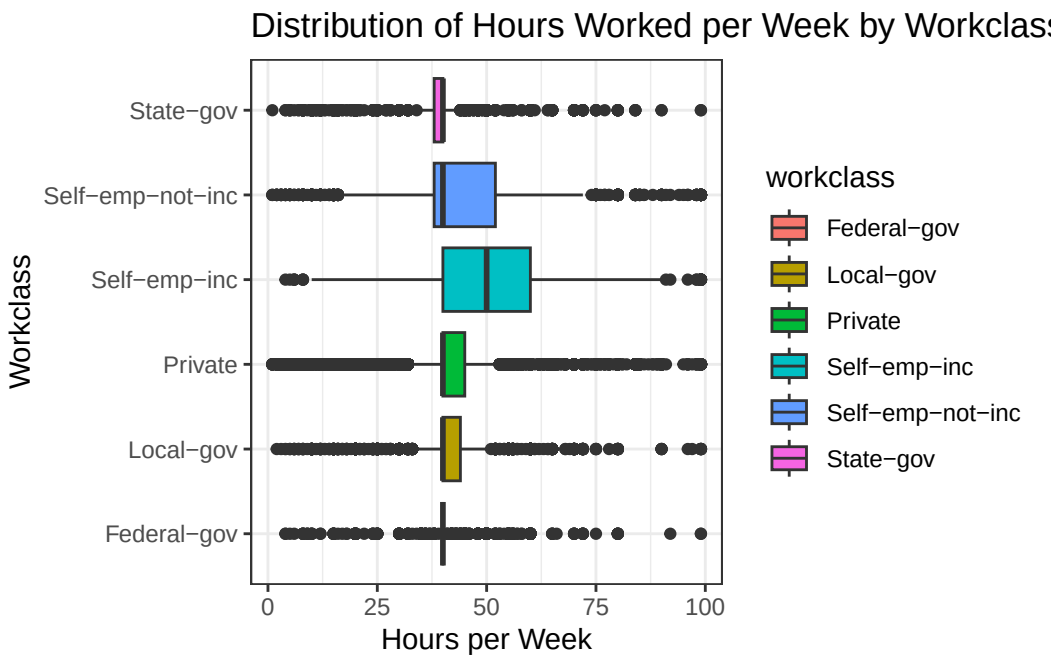
Again, this boxplot shows that age can be a significant factor in income distribution, as older adults are more likely to be in higher income brackets.

Exploring Distribution of Hours worked

```
# Histogram of hours per week
ggplot(Financial_clean, aes(x = hours.per.week)) +
  geom_histogram(binwidth = 1, fill = "blue", color = "black", alpha = 0.7) +
  labs(title = "Distribution of hours per week",
       x = "hours per week",
       y = "Frequency") +
  theme_bw()
```



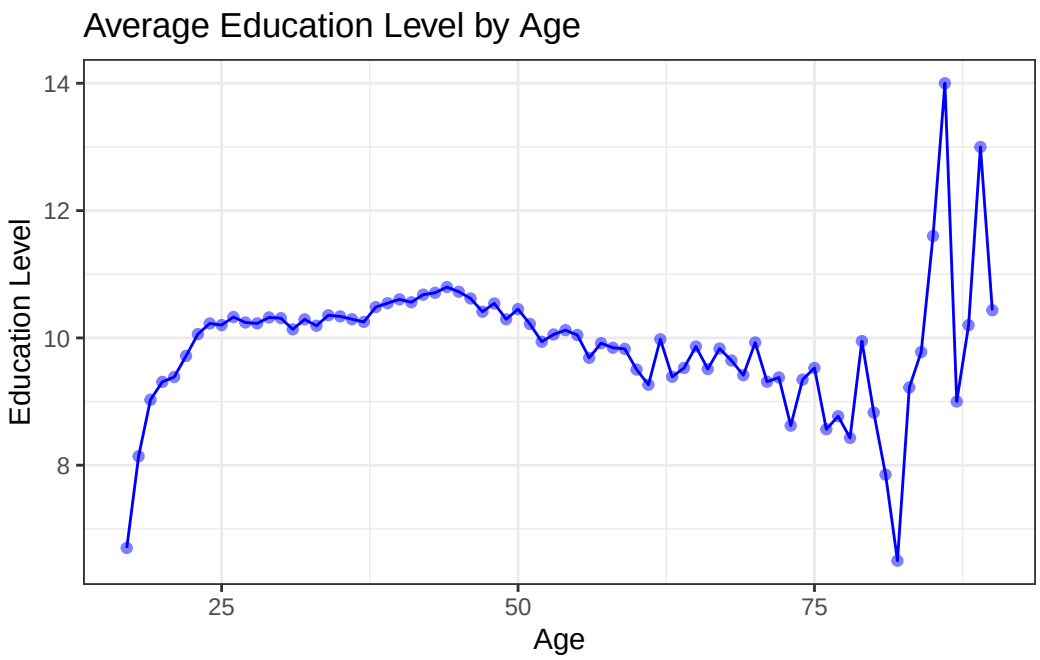

```
#Hours per week distributed by workclass
ggplot(Financial_clean, aes(x = workclass, y = hours.per.week, fill = workclass)) +
  geom_boxplot() +
  coord_flip() +
  labs(title = "Distribution of Hours Worked per Week by Workclass",
       x = "Workclass",
       y = "Hours per Week") +
  theme_bw()
```



The above histogram allows us to examine the overall distribution of hours worked across the dataset, whilst the boxplot provides a more detailed analysis of how these hours per week vary across different work classes. The histogram shows a normal distribution, with majority of hours worked being approximately 36 hours per week. Meanwhile, the boxplot allows us to understand that whilst the lower range of hours across workclasses is consistent, the self-employed workclasses display a larger range of work hours- with self-emp-inc having a noticeably higher median.

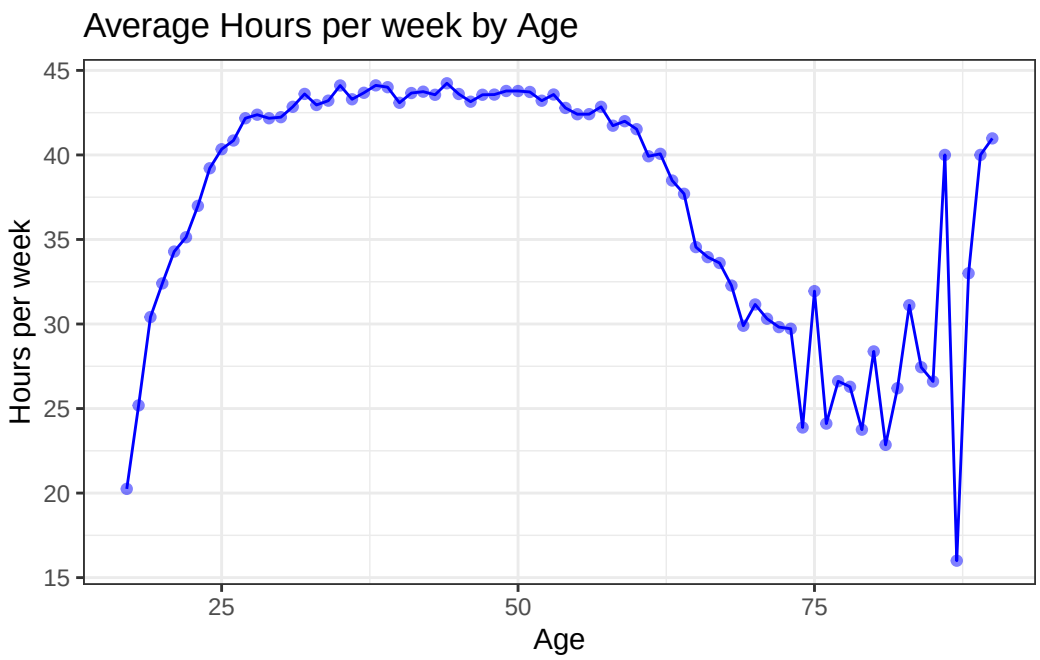
Exploring the age variable

```
# Average Education Level by Age
Financial_clean |>
  group_by(age) |>
  summarise(avg_education_num_by_age = mean(education.num)) |>
  ggplot(aes(x = age, y = avg_education_num_by_age)) +
  geom_point(alpha = 0.5, color = "blue") +
  geom_line(color = "blue") +
  labs(title = "Average Education Level by Age",
       x = "Age",
       y = "Education Level") +
  theme_bw()
```



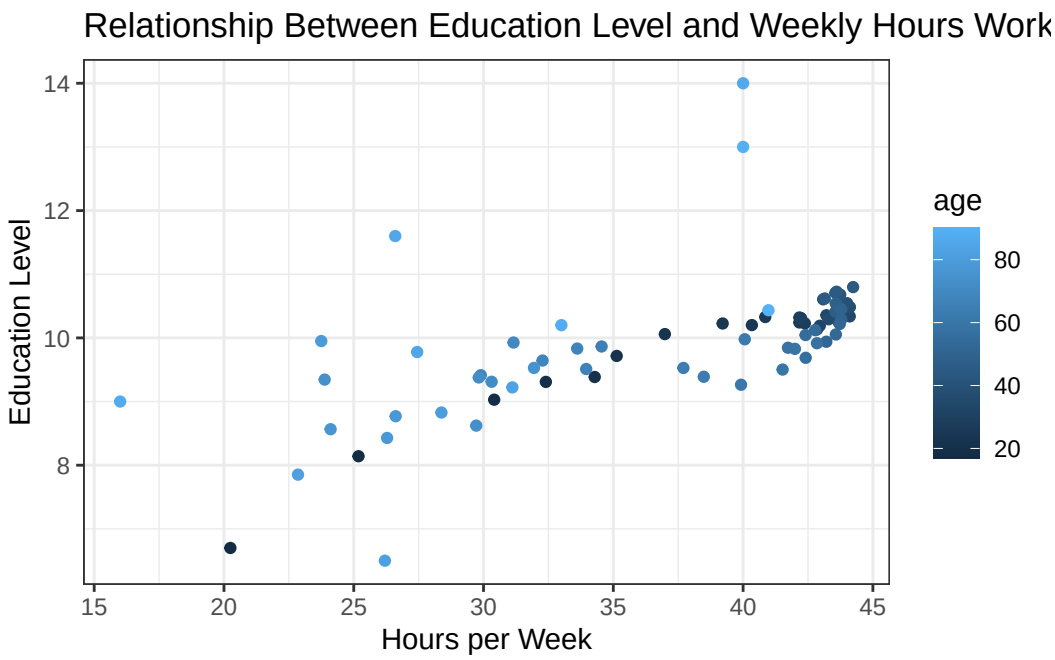
This scatterplot shows how education levels vary with age. The line clearly shows that education increases with age up to the age of 25 before steadying out. However there is a noticeable disparity towards ages over 75, with education levels varying significantly.

```
# Hours per week by age
Financial_clean |>
  group_by(age) |>
  summarise(avg_hours_per_week_age = mean(hours.per.week)) |>
  ggplot(aes(x = age, y = avg_hours_per_week_age)) +
  geom_point(alpha = 0.5, color = "blue") +
  geom_line(color = "blue") +
  labs(title = "Average Hours per week by Age",
       x = "Age",
       y = "Hours per week") +
    theme_bw()
```



This plot provides insight on how working hours vary per age group. It highlights a similar pattern to the previous plot, in which working hours steadily increase up till the age of 25 before plateuing and showing a significant variety after the age of approximately 75.

```
#Education level by hours per week
Financial_clean |>
  group_by(age) |>
  summarise(avg_education_num_by_age = mean(education.num),
            avg_hours_per_week_age = mean(hours.per.week)) |>
  ggplot(aes(x = avg_hours_per_week_age,
             y = avg_education_num_by_age,
             colour = age)) +
  geom_point() +
  labs(title = "Relationship Between Education Level and Weekly Hours Worked",
       x = "Hours per Week",
       y = "Education Level") +
  theme_bw()
```



This plot is helpful for identifying whether there is a correlation between work h ours and education level. There appears to be a small positive trend, showing that people with higher education levels are likely to work more hours. The color gradient of age is useful in identifying the trend over different age groups, and how they engage in the workforce.

Developed Research Questions

1) Does education level significantly impact the number of hours worked per week?

Rationale

The scatterplot of 'Relationship Between Education Level and Weekly Hours Worked' gives us an early insight into correlations between education level and hours worked per week. However more general data science approaches such as ANOVA can be used to further investigate this, and work out significance levels. This has been successful through numerous other studies, with multiple findings suggesting individuals with higher educational attainment are more likely to engage in full-time employment (U.S. Bureau of Labor Statistics, n.d.).

2) How accurately can we predict whether an individual earns >50k or <=50k based on demographic and employment attributes?

Rationale

The EDA and visualisations investigating income using workforce, age, gender, and education level showcase that income varies across demographics. Peer-reviewed material also highlights the potential of different demographics' high predictive value on income, Boll et al., (2017) finding that at the end of careers women accumulate 46.6% less earning than men, and that this also varies throughout life stages. Data modelling classification models such as logistic regression or random forest can therefore potentially be used to predict income. Approaches like this have been used in previous literature regarding Adult Census income level prediction (Chakrabarty & Biswas, 2018).

3) Can we identify groups of workers based on their demographic and employment attributes?

Rationale

Multiple visualisation from our EDA suggest patterns across different variables, for example 'Distribution of Hours Worked per Week by Work class' shows patterns across different job types, and 'Average education level by age' shows that there are correlations between education and age. This helps us identify variables that may be useful for clustering approaches such as K-means or DBSCAN in order to identify and group workers. To reinforce this, variables such as Gender and education level have effectively been used in previous cluster analysis' to group workers. (Jarinto & Ridsomboon, 2023).

Aims and Objectives:

Overall the aims and objectives derived from this EDA to help us answer these three research questions are as follows:

- To investigate the relationship between education level and hours worked per week using statistical techniques such as ANOVA and correlation analysis.
- To explore the factors influencing income levels using statistical tests and develop predictive models for income classification by building and assessing classification models.
- To group individuals into distinct employment clusters using techniques such as K-means or DBSCAN to explore work patterns and demographics.

References

Boll, C., Jahn, M. and Lagemann, A. (2017) 'The gender lifetime earnings gap—exploring gendered pay from the Life Course Perspective', *Journal of Income Distribution®*, pp. 1–53. doi:10.25071/1874-6322.40355.

Chakrabarty, N. and Biswas, S. (2018) 'A statistical approach to adult census income level prediction', *2018 International Conference on Advances in Computing, Communication Control and Networking (ICACCCN)*, pp. 207–212. doi:10.1109/icacccn.2018.8748528.

Dua, D. and Graff, C. (2019) *UCI Machine Learning Repository: Adult Data Set*. University of California, Irvine. Available at: https://archive.ics.uci.edu/ml/datasets/adult

Jarinto, K. and Ridsomboon, L. (2023) ' "clustering by employee personality", Modern Working World Perspectives on work efficiency in the organizations', *WSEAS TRANSACTIONS ON BUSINESS AND ECONOMICS*, 21, pp. 288–298. doi:10.37394/23207.2024.21.26.

Profile of the labor force by educational attainment (no date) *U.S. Bureau of Labor Statistics*. Available at: https://www.bls.gov/spotlight/2017/educational-attainment-of-the-labor-force/?utm_source=chatgpt.com (Accessed: 12 March 2025).